Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Vortex: Turning LLM Serving into SLA-Guaranteed Enterprise Infrastructure			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	R&D /Engineering Services	Other R&D / Engineering-Related Technologies	*Software*	System Integration
Concept of Proposed Technology	*Modern large language model (LLM) services rely on high-end accelerators such as B200 and H100 GPUs, making GPU efficiency and SLA compliance critical for commercial deployment. However, existing LLM serving stacks (e.g., vLLM) primarily optimize for aggregate throughput and lack mechanisms to guarantee Service Level Agreements (SLAs), resulting in unstable latency, missed deadlines, and costly over-provisioning under peak load.* *Vortex is a production-ready SLA orchestration middleware designed to deliver predictable, cost-efficient LLM services for enterprise and industrial applications. Vortex introduces a new execution abstraction that treats GPU capacity as a constrained and monetizable resource, ``token bandwidth'', and allocates it just-in-time based on each request's SLA requirements. Instead of maximizing raw throughput, Vortex ensures that critical requests receive exactly the bandwidth needed to complete on time, while opportunistically filling unused capacity with lower-priority tasks.* *Vortex is designed for Enterprise AI SaaS, Autonomous Agents, and AIOps, where delayed responses can trigger downstream failures and operational risk. By stabilizing end-to-end latency and enforcing SLA-tiered execution, Vortex enables AI service providers to operate 3–5x more concurrent*			

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	workloads on the same GPU infrastructure, significantly improving infrastructure ROI while delivering reliable, commercial-grade AI services.
Rationale for Joint R&D Project	*Enterprise AI services, such as AIOps, automated remediation, and agent-driven business workflows, require strict SLA guarantees to operate reliably at scale. While Korean SMEs are rapidly deploying LLM-based services, existing serving infrastructures lack the ability to translate application-level SLAs into efficient GPU scheduling decisions, leading to unstable latency, workflow failures, and excessive infrastructure costs.* *This joint R&D project is necessary to bridge the gap between cutting-edge scheduling algorithms and production-grade enterprise systems. UIUC contributes foundational expertise in SLA-aware scheduling, large-scale AI systems, and performance modeling, while the Korean SME provides real-world workloads, operational SLAs, and deployment environments that reflect industrial constraints.* *Collaboration is essential to ensure that the proposed technology moves beyond prototype-level validation and becomes a commercially deployable platform. The joint effort enables co-design across algorithmic optimization, system integration, and operational validation, ensuring that the resulting solution delivers measurable GPU cost reduction, SLA compliance, and scalability in real enterprise environments. This partnership accelerates technology transfer and positions Korean SMEs to offer globally competitive, SLA-guaranteed AI services.*
Keywords	1. Generative AI Serving 2. Agentic Workflow Orchestration 3. Infrastructure ROI Optimization 4. GPU Resource Optimization 5. AIOps & Enterprise Automation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

LLM Applications	Real-Time	Direct Use	Content-Based
Code generation	**38.1%**	30.5%	31.4%
Report generation	**39.1%**	36.2%	24.7%
Deep research	38.6%	**47.1%**	14.3%
Real-time translation	36.2%	**39.9%**	23.9%
Batch data processing	15.6%	**49.6%**	34.8%
Reasoning task	28.9%	**47.4%**	23.7%

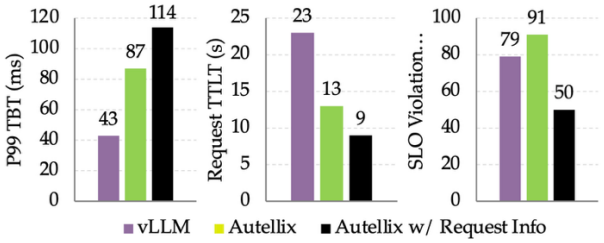

Figure 1: (left) Our real-world user study reports that users exhibit diverse SLA latency needs both across and within applications. "Real-Time" denotes users who prefer low per-token latency; "Direct Use" refers to those demanding for fast full responses; "Content-Based" reflects users whose needs vary depending on the specific context. (right) Existing AI infrastructure managers lack SLA-awareness, resulting in unstable performance, and unpredictable user experience.

Large language models are increasingly deployed as operational agents in enterprise settings, including chat agents, AIOps, and business process automation. Our real-world user studies show that these deployments generate heterogeneous request patterns--latency-sensitive, deadline-sensitive, and compound requests--often within the same workflow and even within a single request's execution lifecycle (Figure 1). These requests exhibit highly variable response lengths, evolving dependencies, and strict end-to-end deadlines, making predictable SLA compliance difficult. Our findings are consistent with recent AI production studies [1]. Existing LLM serving systems [2, 3], including NVIDIA's Dynamo [4], are fundamentally misaligned with these requirements. They optimize aggregate throughput or average latency yet lead to poor application-level SLA satisfaction and service goodput (number of requests meeting SLA latency requirements). As a result, enterprise operators are forced to over-provision GPUs to maintain reliability, leading to low infrastructure efficiency and high operational cost.

The goal of this project is to develop Vortex into a commercial-grade SLA orchestration platform that aligns LLM execution with enterprise SLA requirements while maximizing GPU ROI. Rather than assuming precise request knowledge, Vortex will explicitly embrace uncertainty and dynamically allocate GPU "token bandwidth" to ensure predictable completion of mission-critical workflows.

Current Capabilities (Available Now)

- *Empirical characterization of enterprise LLM workloads, identifying three dominant request patterns and their SLA implications.*
- *Prototype SLA-aware scheduling formulation that maps latency and deadline requirements to token-level bandwidth demands.*
- *Lightweight upper-bound forecasting of response length and basic support for compound (multi-stage) request execution.*
- *Initial integration with vLLM-compatible serving stacks for evaluation under realistic workloads.*

Objectives to Be Achieved via This Joint R&D

- *SLA-aware orchestration engine that enforces per-request and per-workflow SLAs under heterogeneous and bursty enterprise workloads.*
- *Uncertainty-resilient scheduling mechanisms that refine execution decisions online as response lengths and dependencies are revealed.*
- *Compound workflow stabilization, preventing cascading deadline violations in agentic and*

	multi-step automation pipelines. • *SLA-tiered execution support, enabling differentiated service levels (e.g., critical vs. best-effort enterprise tasks).* • *Production-ready integration and validation on the Korean SME's enterprise AI platform, with measurable improvements in SLA compliance, GPU utilization, and operational cost.*
R&D Target Specifications	Vortex will target measurable, system-level outcomes on partner-selected enterprise AI, AIOps, and agentic automation workloads, evaluated on the SME's GPU platforms and serving stack. SLA Compliance and Latency Stability • End-to-end SLA violation rate: ≤ 5% under peak load • P95 Time-to-First-Token (TTFT): ≤ 200 ms, stable across load levels • End-to-end completion for deadline-sensitive workflows: ≥ 95% on-time Infrastructure Efficiency and GPU ROI • GPU operating cost reduction: ≥ 30% versus throughput-optimized baselines • Effective serving capacity: 2.5×–5× more concurrent requests on the same GPU footprint • GPU utilization at equal SLA compliance: ≥ 15–25% improvement Agentic Workflow Reliability • Reduction in cascading failures for multi-step workflows: ≥ 50% • Stable completion of compound/agentic workflows under contention: ≥ 95% success rate Robust Online Adaptation • Detect workload or resource shifts (load spikes, response-length drift): ≤ 30 seconds • Restore SLA compliance for affected traffic: ≤ 2 minutes after disruption
R&D Plan	Phase 1 (0–6 months): Seed-stage planning and prototype alignment • Task 1: Select 2–3 representative enterprise workloads with the SME (e.g., AIOps remediation agent, enterprise RAG assistant, multi-step automation pipeline). • Task 2: Build end-to-end measurement and telemetry harness (latency, SLA, GPU usage). • Task 3: Implement Vortex v0: SLA-aware token-bandwidth model and baseline scheduler integrated with the SME's serving stack. • Deliverables: Workload specifications; KPI definitions; Vortex v0 prototype; Phase 2 execution plan. Phase 2 (6–18 months): Core algorithms and system prototype • Task 4: Develop autonomous, uncertainty-aware scheduling with online refinement of response-length and deadline estimates. • Task 5: Implement agentic workflow stabilization (dependency tracking and stage-level SLA enforcement). • Task 6: Add SLA-tiered scheduling (e.g., critical vs. best-effort workloads) and fairness controls. • Deliverables: Vortex v1 with APIs; experimental evaluation on SME workloads; deployment in staging environment. Phase 3 (18–30 months): Reliability hardening and productization • Task 7: Online adaptation policies for bursty load, hardware variability, and model heterogeneity. • Task 8: Production-like validation: long-running stress tests, failure injection, KPI regression tracking. • Task 9: Deployment toolkit: monitoring hooks, operator onboarding guide, and integration documentation. • Deliverables: Vortex v2 pilot deployment; validation report; commercialization and rollout plan with the SME.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Design and implementation of the core optimization algorithms: • Design and implementation of core SLA-aware orchestration algorithms (token-bandwidth modeling, scheduling, adaptation). • Development and maintenance of the Vortex prototype software stack and APIs. • Performance evaluation, root-cause analysis of SLA violations, and system optimization. • Technical documentation, reference architectures, and deployment guidelines. • Support IP strategy and ensure export-control–compliant technical disclosures.
Expected R&R of Korean SME Partner	Provision of domain-specific data and real-world AI models (the "workload"): • Provide enterprise use cases, SLAs, and success metrics (latency, reliability, cost). • Provide access to representative GPU platforms, serving environments, and operational traces (as permissible). • Integrate telemetry, scheduling hooks, and control interfaces into the existing serving stack. • Lead validation in realistic operational settings, including QA and pilot deployments. • Allocate engineering resources for co-development, integration, and iteration toward commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D will deliver Vortex, a deployable SLA orchestration layer that transforms LLM serving from best-effort execution into a reliable, cost-efficient enterprise infrastructure. • Technological Impact: Establish a state-of-the-art SLA-aware orchestration framework that outperforms existing throughput-driven serving systems in reliability, efficiency, and scalability. • Commercialization Potential: Enable the SME to offer SLA-guaranteed AI services and differentiated enterprise products, directly applicable to AIOps, enterprise automation, smart manufacturing, and mobility. • Economic Impact: Lower the cost barrier for deploying large-scale LLM services by improving GPU ROI, potentially saving hundreds of thousands to millions of dollars annually for enterprise operators. • Long-Term Value: Provide a reusable platform that can continuously onboard new models, agents, and workflows without redesigning infrastructure logic, strengthening the SME's global competitiveness.

6. References

(1) Melissa Z. Pan, Negar Arabzadeh, and et al. Measuring Agents in Production. Arxiv preprint: 2512.04123.

(2) https://github.com/vllm-project/vllm, A high-throughput and memory-efficient inference and serving engine for LLMs.

(3) Michael Luo, Xiaoxiang Shi, Colin Cai, Tianjun Zhang, Justin Wong, Yichuan Wang, Chi Wang, Yanping Huang, Zhifeng Chen, Joseph E. Gonzalez, and Ion Stoica. Autellix: An efficient serving engine for llm agents as general programs. NSDI, 2026.

(4) https://github.com/ai-dynamo/dynamo, NVIDIA Dynamo: A Datacenter Scale Distributed Inference Serving Framework.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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